



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

**Kawayan AI: The Design and Implementation of an Intelligent Content Generation
and Scheduling Platform for Enhanced Philippine Business Visibility**

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EXECUTIVE SUMMARY

This study, entitled “Kawayan AI: The Design and Implementation of an Intelligent Content Generation and Scheduling Platform for Enhanced Philippine Business Visibility,” focuses on the development of an AI-powered platform designed to help Micro, Small, and Medium Enterprises (MSMEs) improve their digital marketing presence. In the Philippines, MSMEs play a major role in employment and economic growth, yet many still struggle to maintain a consistent and professional online presence due to limited time, budget, technical skills, and access to affordable marketing services.

The study addresses key challenges faced by MSMEs, including the high cost of digital marketing services, difficulty in producing culturally relevant content, and lack of efficient tools for planning and scheduling social media posts. While generic AI tools are available, many fail to properly capture the Filipino business context, especially in generating natural and relatable Taglish content. Because of this, Kawayan AI was designed to provide automated, localized, and culturally appropriate content for Philippine MSMEs.

Kawayan AI is a web-based intelligent content generation and scheduling platform that allows users to register their business, submit verification documents, create a Brand DNA profile, generate AI-assisted marketing content, organize posts through a content calendar, and monitor basic performance through a dashboard. The system also includes an admin dashboard for business verification, subscription oversight, user management, and analytics.



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The study uses the Design and Development Research approach, supported by the Iterative Prototyping Model, to guide the design, development, testing, and refinement of the system. It also uses the Technology Acceptance Model to evaluate the platform in terms of perceived usefulness, perceived ease of use, user satisfaction, and system acceptance. The study will involve 40 selected MSME owners, operators, or individuals who manage business social media accounts in Parañaque City.

Overall, Kawayan AI is developed as a practical and affordable digital marketing support tool for MSMEs. By combining AI-driven content generation, culturally intelligent prompting, scheduling support, analytics, and business verification, the platform aims to help local businesses improve their digital visibility and become more competitive in the modern digital marketplace.



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CHAPTER 1

THE PROBLEM AND ITS SETTING

Introduction

The vitality of emerging economies is inextricably linked to the performance, resilience, and scalability of Micro, Small, and Medium Enterprises (MSMEs). In the context of the Republic of the Philippines, this sector does not merely represent a fraction of the commercial market; it constitutes the foundational pillar of the nation's economic activity. Statistical data reveals that MSMEs account for an astounding 99.63 percent of all registered business establishments in the country, totaling over 1.2 million enterprises. Furthermore, this sector functions as the primary driver of national employment, providing livelihoods for over 62.4 percent of the Filipino workforce and contributing approximately 36 percent of the gross value added. Consequently, the competitive viability of MSMEs exerts an immediate, profound, and systemic influence on the broader socioeconomic structure. Any technological intervention designed to enhance the operational efficiency and market reach of these enterprises must, therefore, be viewed not merely as a commercial product, but as a strategic macroeconomic mechanism that directly supports national employment targets, long-term economic stability, and poverty alleviation.

The contemporary operational environment for these enterprises is characterized by an escalating degree of digital saturation, alongside compounding global crises. The 2025 United Nations Sustainable Development Goals (SDG) Report indicates that the world is facing a global development emergency, with only 35 percent of SDG targets currently on track or making moderate progress. Nearly half of the targets are moving



too slowly, and 18 percent are actively reversing due to interconnected challenges such as climate chaos, geopolitical tensions, and economic shocks. Over 800 million people globally remain trapped in extreme poverty, while debt servicing costs in developing nations squeeze the resources necessary for sustainable development. Within this turbulent landscape, described by the International Council for Small Business as a state of "Permanent White Water", disruption is no longer episodic but constant, requiring MSMEs to navigate rapid digital transformation and supply chain volatility with fewer buffers and greater exposure to risk.

The Philippines is widely acknowledged as a hyper-digitalized market, boasting over 86.75 million active social media users, which equates to approximately 73.4 percent of the total population. This high internet penetration rate solidifies digital platforms as the central, indispensable channels for commerce, customer engagement, and brand building. Consequently, maintaining a consistent, high-quality, and strategically aligned online presence has transitioned from a supplementary marketing tactic to a mandatory requirement for business survival. However, a critical systemic paradox exists: despite the overwhelming economic significance of MSMEs and the undeniable necessity of digital engagement, approximately 75 percent of Filipino businesses remain digitally marginalized.

This marginalization is predominantly driven by severe institutional and resource constraints, including limitations on time, budget, and access to the technological infrastructure required to produce consistent, high-quality digital content. Professional digital marketing services in the Philippines typically command monthly retainer fees ranging from PHP 25,000 to PHP 200,000, a financial burden that is entirely unsustainable for micro and small enterprises. The advent of Artificial Intelligence (AI),



particularly generative large language models (LLMs), presents an unprecedented opportunity to resolve this resource disparity. Systems designed to automate the ideation, generation, and scheduling of culturally resonant content, such as the Kawayan AI platform, serve as critical equalizers in the digital economy. By leveraging advanced batch-processing algorithms, localized cultural prompting frameworks, and low-cost subscription models, such technologies dismantle the financial and technical barriers that have historically excluded MSMEs from the digital marketplace.

To fully understand the macroeconomic utility of such technological systems, it is necessary to rigorously map their functional architectures against globally recognized development frameworks. The United Nations Sustainable Development Goals, adopted in 2015 as a universal call to action to end poverty, protect the planet, and ensure prosperity, provide a comprehensive rubric for this analysis. Concurrently, national strategic blueprints, specifically the Philippine Development Plan (PDP) 2023-2028, articulate the localized pathways through which these global goals are to be achieved. This study provides an exhaustive, multi-dimensional analysis of how intelligent, localized content generation platforms systematically align with, and actively accelerate, specific UN SDGs and national development targets.

Sustainable Development Goals (SDGs) Alignment

SDG 8: Decent Work and Economic Growth

Sustainable Development Goal 8 advocates for sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all. For developing economies, the realization of this goal is fundamentally dependent on the modernization and capacity-building of the MSME sector. The deployment of automated, AI-driven digital marketing platforms intersects directly with several critical targets under



SDG 8, primarily by reducing operational friction, combating informal labor dynamics, and enabling micro-enterprises to compete in high-value digital markets.

Target 8.2: Economic Productivity and Technological Upgrading

Target 8.2 mandates the achievement of higher levels of economic productivity through diversification, technological upgrading, and innovation, with a specific focus on high-value-added and labor-intensive sectors. Historically, MSMEs in the Philippines have suffered from stagnant productivity levels due to a reliance on manual, time-intensive operational processes. Business owners frequently assume multifarious roles, acting simultaneously as product developers, customer service representatives, financial controllers, and marketing managers. This fragmentation of attention severely limits the cognitive bandwidth available for strategic growth and innovation.

Intelligent content generation platforms address this productivity bottleneck through extreme workflow automation. By utilizing advanced LLMs, such as the Google Gemini engine, to execute "Batch Ideation Generation," a platform can synthesize a comprehensive, culturally calibrated, month-long content calendar comprising 8 to 16 targeted posts in a single computational transaction. This technological upgrading shifts the MSME owner's role from a manual content creator to a strategic editor. The time and labor previously expended on daily social media management can be reallocated to core productive activities, such as product quality enhancement, supply chain optimization, and direct customer relations. The resulting increase in operational efficiency directly satisfies the mandate of Target 8.2, proving that algorithmic



automation is a viable pathway to higher aggregate economic productivity without necessitating corresponding increases in human labor hours.

Target 8.3: Promoting Formalization and Growth of MSMEs

Target 8.3 explicitly calls for the promotion of development-oriented policies that support productive activities, entrepreneurship, creativity, and innovation, and encourages the formalization and growth of micro, small, and medium-sized enterprises. The informal sector in the Philippines is vast, with estimates suggesting that up to 70 percent of the workforce engages in informal, unprotected labor. Globally, the data is similarly stark, indicating that in 2024, 57.8 percent of the global workforce was in informal employment, lacking social protection—an increase of 34 million workers compared to the previous year. Informal MSMEs often face insurmountable hurdles when attempting to scale, primarily due to their inability to access formal credit markets, secure business-to-business (B2B) contracts, or establish verifiable brand credibility.

A consistent, professional digital footprint serves as a modern proxy for business legitimacy. When an MSME utilizes an AI platform to maintain a highly professional, culturally resonant online presence, it establishes "Perceived Brand Localness" (PBL) and structural brand credibility. This digital visibility is frequently a prerequisite for formalization. Financial institutions, investors, and larger supply-chain partners increasingly rely on digital presence and social media metrics as indicators of operational stability and market traction during due diligence processes. By dramatically lowering the cost of acquiring this digital legitimacy—from prohibitive agency fees to accessible Software-as-a-Service (SaaS) micro-subscriptions—technology platforms indirectly incentivize and



facilitate the transition of enterprises from the informal, localized shadow economy into the formalized digital marketplace.

Target 8.10: Universal Access to Financial Services

Target 8.10 focuses on strengthening the capacity of domestic financial institutions to encourage and expand access to banking, insurance, and financial services for all. MSMEs often struggle to access affordable financing options because financial institutions perceive them as risky borrowers due to their small size, lack of collateral, and limited financial track records. MSMEs frequently avoid commercial banks due to high-interest rates, voluminous requirements, and short repayment periods.

The integration of automated financial mechanisms within digital platforms bridges this gap. Platforms like Kawayan AI implement localized payment gateways (e.g., the Xendit SDK) that support recurring billing via local e-wallets such as GCash and Maya, alongside traditional banking methods. This architecture not only accommodates the unbanked and underbanked populations but also establishes a recorded, verifiable financial history for the subscribing MSME. Over time, this digital payment footprint serves as alternative credit data, allowing MSMEs to demonstrate financial reliability. This technological enablement aligns seamlessly with the objectives of Target 8.10, integrating previously invisible micro-enterprises into the formal financial service ecosystem.

SDG 9: Industry, Innovation, and Infrastructure

Sustainable Development Goal 9 focuses on building resilient infrastructure, promoting inclusive and sustainable industrialization, and fostering innovation. While traditional interpretations of this goal often center on physical infrastructure such as



transport networks and manufacturing plants, the modern economic paradigm dictates that digital infrastructure and computational innovation are equally critical to sustainable industrialization and economic resilience.

Target 9.b: Domestic Technology Development and Cultural Innovation

Target 9.b urges support for domestic technology development, research, and innovation in developing countries, ensuring a conducive policy environment for industrial diversification and value addition. The design and implementation of culturally intelligent AI systems represent a profound advancement in domestic technological research. Generalized, Western-centric AI models inherently struggle with the nuances of low-resource languages and complex socio-linguistic phenomena, such as the Taglish code-switching prevalent in the Philippines. When applied generically, these models produce outputs that suffer from "Linguistic Negative Transfer," resulting in content that feels robotic, culturally sterile, awkwardly phrased, or inadvertently offensive to the target demographic.

Research and development efforts that bridge this gap—such as the creation of a specialized "Culturally Intelligent Prompting Framework"—constitute vital domestic innovation. By implementing a "chain of translations" methodology, where the AI is engineered to process semantic meaning in high-resource languages before executing culturally nuanced translations into localized dialects, developers generate highly authentic outputs that resonate with the local consumer base. This architectural innovation ensures that the economic benefits of generative AI are not restricted to English-speaking, high-income markets, but are localized to empower the domestic economy. This fulfills the mandate of



Target 9.b by fostering technological capabilities that are highly customized to national circumstances, directly driving the "Perceived Brand Localness" that correlates with consumer trust and purchase intentions.

Target 9.3: Integration of Small-Scale Enterprises into Value Chains

Target 9.3 specifically mandates increasing the access of small-scale industrial and other enterprises to financial services and their integration into broader value chains and markets. The digital divide is a primary factor isolating MSMEs from national and global value chains. Larger corporations mandate that their suppliers, distributors, and retail partners maintain sophisticated, transparent, and digitally integrated operations. Without the requisite digital knowledge, human resources, or financial capacity to invest in these systems, MSMEs risk their competitiveness and integration into larger regional or global value chains.

When an MSME utilizes an automated platform to maintain a robust digital presence and track consumer engagement metrics, it fundamentally upgrades its operational capacity. Systems that integrate tools like the bundle.social API for secure cross-platform synchronization and provide Growth Insights dashboards for tracking reach and engagement transform an isolated micro-business into a data-driven entity. By acting as a digital bridge, the platform generates the performance analytics required to negotiate with larger value chain actors, facilitating the integration of isolated micro-businesses into the broader supply chain ecosystem, thereby directly advancing the objectives of Target 9.3.

Target 9.c: Affordable ICT Access and Resilient Infrastructure



Target 9.c aims to significantly increase access to information and communications technology (ICT) and strive to provide universal and affordable access to the internet in developing countries. A persistent barrier to the adoption of advanced digital tools in regions like the Philippines is the inconsistency of internet infrastructure, particularly outside major urban centers. The software architecture of equitable AI platforms must actively account for these infrastructural deficiencies. Systems that rely on heavy, constant server-side rendering or massive data payloads are fundamentally incompatible with low-bandwidth environments.

Platforms designed for the MSME context must employ "low-bandwidth interfaces" and asynchronous processing mechanisms. The Kawayan AI architecture achieves this by allowing users to generate a full month's worth of content via a single, lightweight API call, minimizing the need for constant, stable connectivity during the ideation phase. Furthermore, the utilization of serverless, zero-configuration database engines like SQLite facilitates high-performance, multi-tenant data storage without the latency and cost associated with remote cloud database queries over unstable connections. The physical architecture also utilizes a Flat-File Storage System, keeping high-resolution AI-generated graphics in protected directories rather than bloating the database with Binary Large Objects (BLOBS). This architectural frugality ensures that the platform remains highly responsive and accessible even on standard, low-speed mobile internet connections, thereby actively supporting the ethos of Target 9.c by making advanced ICT practically usable for rural and technologically constrained populations.



Theoretical Framework

The adoption and diffusion of artificial intelligence within the micro, small, and medium enterprise sector can be rigorously analyzed through the Technology-Organization-Environment (TOE) framework and the Diffusion of Innovations (DOI) theory. The TOE framework is essential for evaluating how technological capabilities, organizational readiness, and environmental market dynamics dictate the successful adoption and execution of new IT systems in businesses. Concurrently, the DOI theory explains how these innovations spread, are perceived, and ultimately accepted within a competitive ecosystem.

Figure 1. Design and Development Research (DDR) Cycle

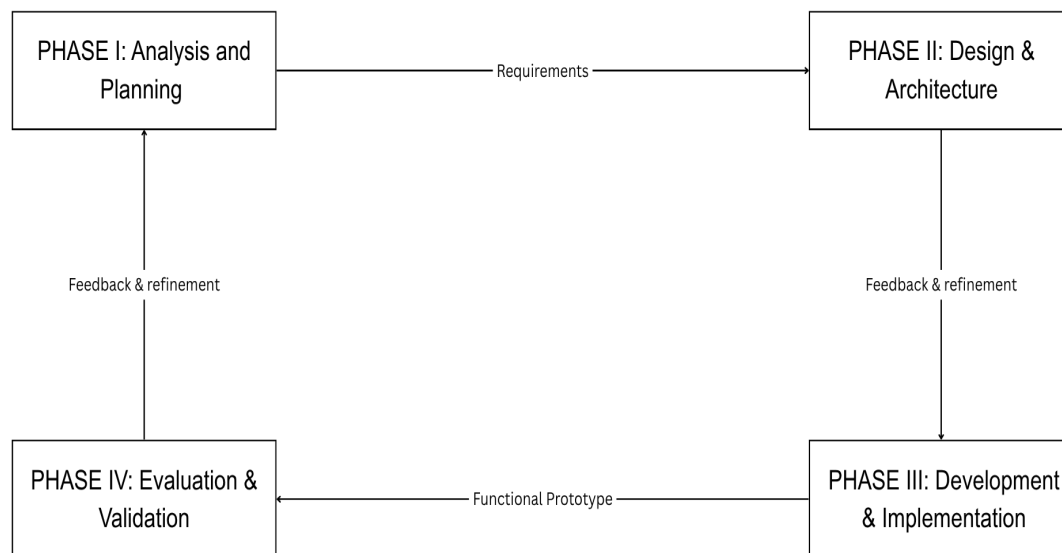


Figure 1 illustrates the iterative cycle of the Design and Development Research (DDR) paradigm that anchors the Kawayan AI study. This approach represents a systematic study of development, design, and evaluation processes, establishing an

empirical basis for the creation of the intelligent content generation platform. The cycle begins with Phase I (Analysis and Planning) to identify the specific digital marginalization and resource constraints of Philippine MSMEs. It then proceeds through a structured Design and Architecture phase and a Development phase, before concluding with Evaluation and Validation to verify the platform's effectiveness in enhancing business visibility.

In the context of localized digital marketing for Philippine MSMEs, the platform's cultural localization capabilities are profoundly anchored in the theory of Perceived Brand Localness (PBL). PBL asserts that when a brand is recognized as an authentic, local iconic, or cultural symbol, it significantly enhances brand credibility, emotional brand love, and consumers' booking or purchase intentions. By systematically guiding AI to generate culturally resonant content, the platform actively builds this perception of localness, strengthening the brand's position and trust within the domestic market.

Figure 2. Adapted Technology Acceptance Model (TAM)

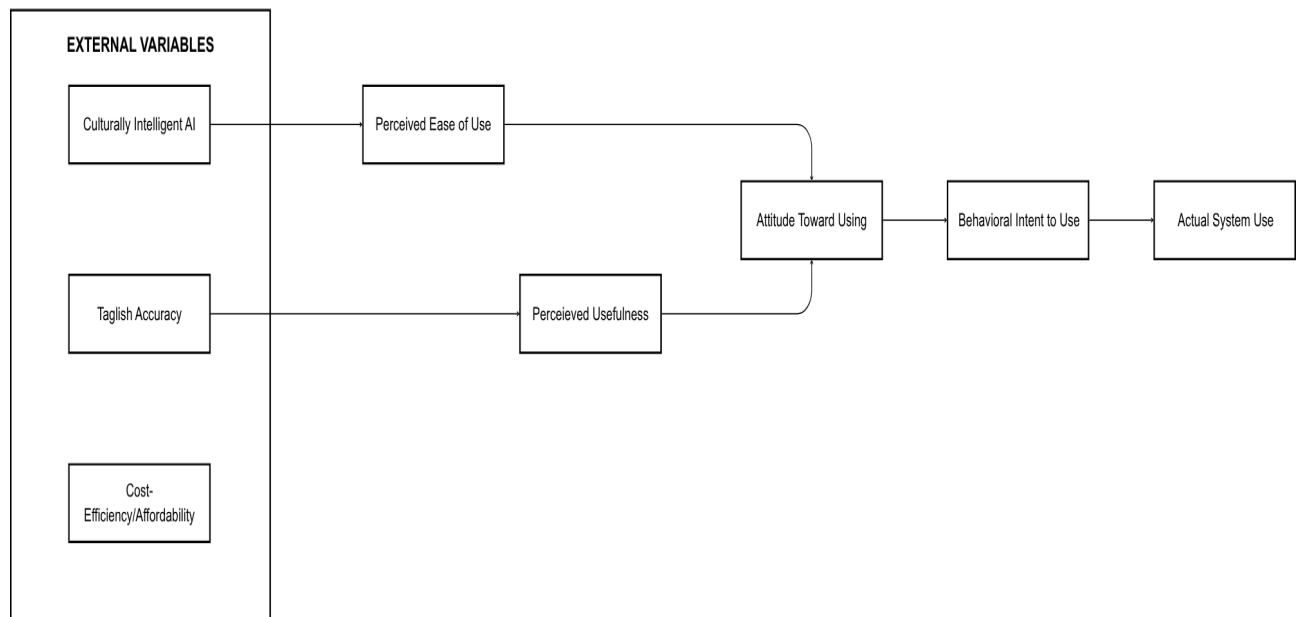




Figure 2 illustrates the adapted Technology Acceptance Model (TAM) used to evaluate user adoption and the long-term sustainability of the Kawayan AI platform. The model identifies critical external variables, specifically Culturally Intelligent AI and Taglish Accuracy, which serve as the primary drivers for Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). In this context, the system's ability to produce authentic, localized content at a low cost directly influences the MSME owner's attitude toward the technology, which dictates their behavioral intention and eventual Actual System Use.

Perceived Usefulness (PU): This measures the degree to which an MSME owner believes that utilizing Kawayan AI will significantly enhance their marketing productivity and competitive visibility. In the context of this study, PU is validated by the system's ability to reduce content ideation cycles from days to minutes through Batch Ideation Generation, while ensuring that the resulting "Taglish" content achieves a high degree of Perceived Brand Localness (PBL). High PU is achieved when the user is assured that the platform's outputs are not merely automated, but are culturally authentic enough to foster consumer trust and purchase intention.

Perceived Ease of Use (PEOU): This measures the user's belief that interacting with the system will be effortless and intuitive, requiring minimal cognitive load. For MSMEs with varying levels of digital literacy, PEOU is essential in minimizing adoption resistance. This is addressed through the platform's Goal-Based Navigation, which allows users to generate month-long calendars without managing complex technical parameters, and the low-bandwidth interface designed for the Philippine infrastructural reality.



Furthermore, the study integrates a Strategic Engagement Loop via the Growth Insights Dashboard. This feature provides non-monetary "performance rewards" by translating engagement metrics into visual ROI charts, which serve to reinforce the system's perceived usefulness. By demonstrating tangible business growth, the dashboard provides the social reinforcement necessary for high user retention and sustained content consistency, which are crucial for the long-term sustainability of the software.

By adopting DDR and adapting TAM, the study ensures that Kawayan AI is not only built correctly but is also proven to be a practical, efficient, and culturally aligned solution for its target user community.

Conceptual Framework

The study utilizes an Input-Process-Output (IPO) model to map the operational flow of the automated marketing platform:

Figure 3. Input-Process-Output Model

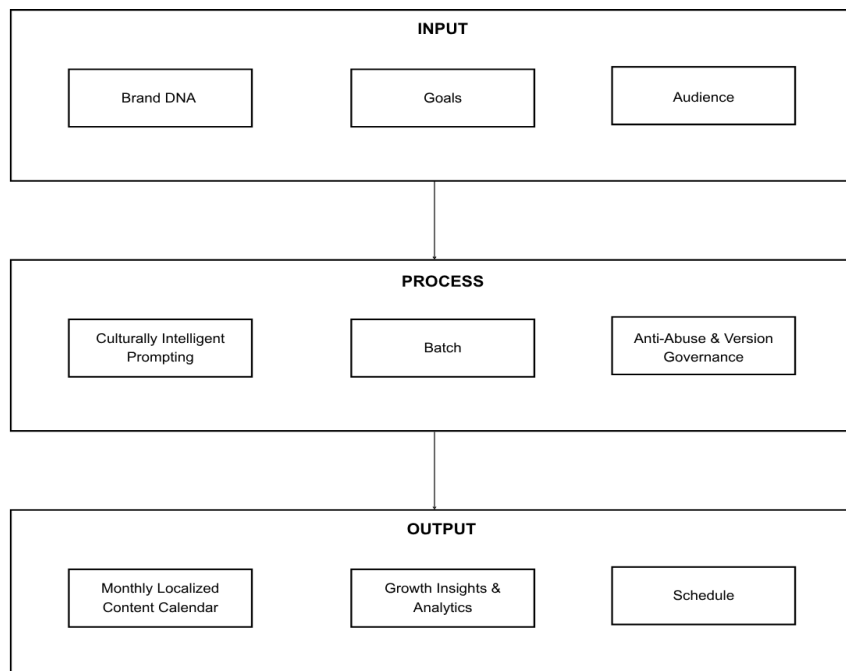




Figure 3 illustrates the flow of the **Input-Process-Output (IPO)** model, which defines the operational boundaries and functional scope of the Kawayan AI platform.

- **Input:** The system ingests **Brand DNA** (voice, audience demographics, and goals) and visual identity parameters (HEX codes) provided by the MSME owner through the strategic brand survey.
- **Process:** These inputs are transformed through the **Culturally Intelligent Prompting Framework** and the **Google Gemini API**, utilizing "Batch Ideation Generation" and a "Chain of Translations" to ensure linguistic and cultural nuance.
- **Output:** The final deliverables include a **Monthly Localized Content Calendar** (consisting of 8 or 16 posts), automated scheduling data, and an interactive **Growth Insights** dashboard for engagement analytics.

Iterative Prototyping Model (IPM) Phases

The software development life cycle (SDLC) for the platform is governed by the **Iterative Prototyping Model (IPM)**. This model ensures that the system is developed in small, manageable sections, allowing for continuous refinement and adaptation based on user evaluation. The IPM cycle for Kawayan AI is structured into four distinct phases:

- **Phase 1: Planning and Requirements Analysis:** Identifying the specific financial constraints, digital literacy gaps, and infrastructural realities of Philippine MSMEs. This involves defining both functional requirements, such as **Batch Ideation Generation**, and non-functional requirements, including high performance on **low-bandwidth connections**.
- **Phase 2: System Design:** Architecting the foundational **MERN stack**, designing the **multi-tenant SQLite database schema** for logical data isolation, and



establishing a **flat-file storage system** to ensure the database remains lightweight and responsive for content scheduling.

- **Phase 3: Development and Implementation:** Constructing the functional code for all core modules, including the integration of the **Google Gemini API** for natural language reasoning, the **bundle.social API** for cross-platform synchronization, and the **Xendit SDK** for localized recurring payments via GCash and Maya.
- **Phase 4: Testing and Evaluation:** Conducting rigorous **User Acceptance Testing (UAT)** to evaluate **Taglish localization** accuracy and refining the "**Chain of Translations**" methodology. This phase identifies and resolves "**Linguistic Negative Transfer**," ensuring the system delivers culturally authentic content that achieves **Perceived Brand Localness (PBL)**.

Statement of the Problem

The digital transformation of the Philippine economy has reached a crisis degree of disparity, where Micro, Small, and Medium Enterprises (MSMEs) are increasingly marginalized despite the nation's high social media penetration. While there are generic tools available, the absence of an affordable and culturally calibrated framework significantly restricts their competitive efforts. These groups' key major issues are as follows:

1. **Prohibitive Operational Costs:** The existing method for acquiring professional digital marketing services is financially inaccessible for MSMEs. Monthly retainer fees typically command between PHP 25,000 to PHP 200,000, a burden that is entirely unsustainable for micro-enterprises. This leads to a systemic exclusion of small businesses from high-value digital marketplaces.



2. **Lack of Cultural Localization and Nuance:** General-purpose AI models are unable to meet the specific linguistic and cultural requirements of the Philippine market. Rescuing a brand's digital identity without a localized system leads to awkward "Taglish" code-switching or cultural mistakes that alienate 24% of consumers. The absence of a culturally intelligent system prevents businesses from attaining "Perceived Brand Localness," which is critical for building consumer trust.
3. **Inefficient Time and Resource Management:** MSME owners are unable to maintain consistent online visibility due to severe time poverty and the absence of automated workflows. Most enterprises manage social media reactively, lacking the bandwidth for strategic planning. The absence of a centralized, batch-processing system makes it impossible to monitor a content strategy from initial ideation to final scheduling, resulting in wasted human resources and lost market opportunity.

By addressing these problems, this study aims to provide a technology-driven solution, the Kawayan AI platform, that empowers MSMEs through automated, culturally intelligent content generation, and ultimately contributes to a more inclusive and effective digital economy in the Philippines.

Scope and Limitations of the Study

This study is focused on designing and developing a culturally intelligent content generation and scheduling system to address the digital marketing challenges faced by MSMEs in Parañaque City, Philippines. The study's scope includes the following:



Scope of the Study

- **Geographic Area:** The system will be pilot tested exclusively within Parañaque City, Philippines. The findings and results will be specific to this location
- **Target Participants:** The study will involve local MSME owners and a designated partner organization within the Parañaque pilot area.
- **Core Functionality:** The system will implement and evaluate core features into three primary sections (The Client Interface, the Automation Core, and the Admin Dashboard).

1. **Client Interface (Web Browser):** Clients will be able to utilize the system for Brand DNA creation, automated content production, and engagement monitoring.

a. Business Registration and Verification

- i. **Business Information Submission:** The system allows MSME users to register by providing basic business information, including business name, email address, contact number, and business address.
- ii. **Document Upload:** To verify business legitimacy, users are required to upload at least one valid business registration document, such as a Mayor's Permit or DTI/SEC Registration.
- iii. **Verification Status Display:** After submitting the required information and document, the system displays the user's verification status as "Verification Pending" while the administrator reviews the submitted document.
- iv. **Restricted Access Before Approval:** Users with pending or rejected verification status may have limited access to platform



features until their business document is reviewed and approved by the administrator.

b. Strategic Brand Survey

- i. **Digital Brand Persona Capture:** Collects essential data such as brand voice, target audience, and preferred cultural tone to build a "Digital Brand Persona".
- ii. **Visual Identity Mapping:** Requires the user to input brand colors (HEX codes) to ensure aesthetic consistency in generated prompts.

c. Batch AI Content Generation

- i. **Goal-Based Prompting:** A simple interface where users input a specific marketing goal (e.g., "Holiday Sale Promotion").
- ii. **Batch Ideation Engine:** Simultaneously generates a full month's content (8 posts for Trial, 16 for Pro) in a single operation.
- iii. **Linguistic Taglish Optimization:** Employs a "chain of translations" to ensure captions resonate with local consumers through culturally accurate Taglish code-switching.

d. Management and Performance

- i. **Interactive Editorial Calendar:** A central hub to visualize the monthly schedule, featuring an "Add-on" module for supplemental posts.
- ii. **Growth Insights Dashboard:** Translates engagement metrics into visual charts to demonstrate digital Return on Investment (ROI).



- iii. **Anti-Abuse Safeguards:** Enforces a strict two-attempt Regeneration Limit and utilizes "Version Obfuscation" (blurring) to prevent resource exploitation.
- 2. **Automation Core (System Engine):** The underlying engine responsible for data flow and cross-platform communication.
 - a. **Culturally Intelligent Prompting Framework:** Guides the commercial LLM (Google Gemini) to produce authentic, localized outputs.
 - b. **Automated Scheduling Synchronization:** Uses the bundle.social API to facilitate post preparation and cross-platform synchronization.
 - c. **Localized Payment Processing:** Integrates the Xendit SDK to handle recurring subscription billing via local channels like GCash and Maya
- 3. **Admin Dashboard (Web Oversight):** This interface is used by system operators for platform oversight and performance evaluation.
 - a. **Verification & Control**
 - i. **Business Verification Queue:** A centralized admin interface where system operators can review newly registered MSME accounts. The dashboard displays the business name, uploaded document, and current verification status. The administrator may open the uploaded document, check if it appears legitimate and matches the submitted business name, then approve or reject the registration.
 - ii. **Verification Result Management:** Once the administrator approves or rejects the submission, the system will automatically update the verification status of users. Approved users are marked



as "Verified Business" and will be able to access the platform's full features, while rejected users are marked as "Verification Failed" and are instructed to upload another business registration document for validation.

- iii. **Subscription Oversight:** Tools to verify payment status and manage active billing cycles using Xendit webhooks.

b. Data & Evaluation

- i. **System-Wide Analytics:** KPI visualization tracking aggregate user growth, retention rates, and platform-wide content performance.
- ii. **Integrated Help Desk:** A ticketing system (starting at ID 1001) for managing technical and billing concerns from clients.

Limitations of the Study

1. **Technological Dependency:** The system's effectiveness is contingent on users having reliable internet connectivity. While a "low-bandwidth interface" is provided, zero connectivity prevents content generation.
2. **Platform Exclusivity:** The system is developed as a web-based application, native mobile applications for iOS or Android are outside the current scope.
3. **Exclusion of Advanced Features:** The study does not include video content creation (e.g., TikTok/Reels) or direct native publishing. It relies on an external commercial API (Google Gemini) rather than a proprietary foundational model.
4. **Manual Document Verification:** The business verification process is limited to manual review of business documents uploaded by users. The system is not directly connected to government databases such as DTI, SEC, or local



government permit systems. Therefore, verification is based only on the document's visible legitimacy and consistency with the registered business information uploaded.

Significance of the Study

The development of this Kawayan AI platform holds significant importance, with a clear hierarchy of benefits that ultimately serve the primary goal of improving the effectiveness of MSME digital marketing efforts.

To **MSMEs**. The micro, small, and medium enterprises in the Philippines stand to gain the most from this study. With the system's help, their main operating issues are directly addressed. This will help them streamline the entire marketing process by providing them a more dependable, faster, and data-driven approach to their brand presence. The system saves time and money by providing a centralized platform, which reduces the need for costly external agencies that charge up to PHP 200,000 per month. By increasing Perceived Brand Localness (PBL), the system builds consumer trust and prevents the 24% drop in purchase interest associated with cultural inaccuracies.

To **the Public**. The system will provide a structured and culturally relevant digital experience for consumers. By ensuring that local businesses can project an authentic, localized digital brand, the public gains access to more relatable products and services. The democratization of AI ensures that even micro-enterprises can compete with large corporations, providing the public with a wider variety of choice in the digital marketplace and reducing the risk of a "greater global economic divide" where only large firms possess digital voices.

To **the Community**. The system will contribute to a more resilient and inclusive community by enhancing the competitiveness of local businesses that employ 62.4% of



the national workforce. A more organized and responsive digital economy leads to sustained job creation and a decrease in business failures. Furthermore, the study aligns with the Philippine Development Plan (PDP) 2023-2028 Chapters 8 and 10, fostering a dynamic innovation ecosystem and promoting healthy market competition. This collective empowerment strengthens national economic stability and ensures that the digital transformation of the Philippines leaves no entrepreneur behind.

Definition of Terms

This section provides definitions of terms used in this study for better understanding.

1. **Batch Ideation Generation:** A computational mechanism where an AI model synthesizes and generates an entire month's worth of content (8 to 16 posts) through a single API transaction, bypassing manual turn-by-turn prompting.
2. **Chain of Translations:** A localized prompting methodology where the AI is engineered to process semantic meaning in a high-resource language (like English) before executing nuanced translations into domestic dialects, mitigating unnatural phrasing.
3. **Iterative Prototyping Model (IPM):** A cyclical software development methodology where an application is planned, designed, coded, tested, and continuously refined in small subsets or iterations based on stakeholder feedback.
4. **Large Language Model (LLM):** A type of artificial intelligence trained on massive datasets to understand and generate human-like text, serving as the underlying intelligence engine for generating marketing captions.
5. **Micro, Small, and Medium Enterprises (MSMEs):** Businesses in the Philippines classified by asset size (e.g., PHP 3M or less for micro) and



employee count (e.g., 1-9 for micro), forming the backbone of the national economy.

6. **Perceived Brand Localness (PBL):** A marketing theory measuring the degree to which consumers perceive a brand as an authentic, relatable, and deeply integrated cultural symbol within their local market.
7. **Role-Based Access Control (RBAC):** A security strategy that manages user permissions by assigning access rights based on organizational roles (e.g., Client, Admin, Support), ensuring sensitive business data remains protected.
8. **Software-as-a-Service (SaaS):** A cloud-based software delivery model where users access applications via a web browser under a subscription plan, eliminating the need for complex local infrastructure.
9. **Taglish:** A linguistically complex code-switching phenomenon in the Philippines involving the fusion of Tagalog and English grammar, which requires specialized AI prompting to replicate authentically.
10. **Version Obfuscation:** A preventative user-experience feature that visually blurs previously generated content iterations within the platform to enforce decisive action and curb computational resource abuse.



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CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter provides a multidimensional review of the existing literature and studies relevant to the development of Kawayan AI. The discussion is organized into five thematic clusters: the macroeconomic landscape of Philippine MSMEs, the barriers of resource constraints and digital marginalization, the sociolinguistic complexity of Taglish in the digital space, the theoretical foundations of Perceived Brand Localness (PBL), and the role of frontier-class artificial intelligence in bridging the digital divide.

The Philippine Digital Economy and the MSME Pillar

The vitality of the Philippine economy is inextricably linked to the performance of its Micro, Small, and Medium Enterprises (MSMEs). According to 2024 data from the Department of Trade and Industry (DTI), MSMEs constitute 99.63% of all registered business establishments, totaling approximately 1,236,908 enterprises. This sector serves as the nation's primary employment engine, supporting over 62.4% of the total workforce and contributing 36% of the national gross value added.

The growth of these enterprises is occurring within a hyper-digitalized environment. Recent reports by Google, Temasek, and Bain & Co. (2025) project that the Philippine digital economy will reach \$36 billion by 2025, driven largely by e-commerce and digital services. DataReportal (2024) further quantifies this saturation, noting that there are 86.75 million active social media users in the Philippines, representing 73.4% of the total population. This high internet penetration has transformed digital platforms from optional marketing channels into mandatory



infrastructure for business survival. As Dentsu (2025) observes, Filipino consumers are now more "empowered and values-driven than ever before," requiring brands to maintain a consistent and culturally relevant digital presence to remain competitive.

Digital Marginalization and Resource Constraints

Despite the overwhelming necessity of digital engagement, a systemic "digital divide" persists. Entrepreneurship.org.ph (2024) highlights that approximately 75% of Filipino businesses remain digitally marginalized, unable to effectively leverage social media for growth. This marginalization is rooted in severe Resource Constraints. As defined by Ailleron (2024), these constraints encompass limitations in manpower, technology, time, and budget.

Wazile (2024) provides a granular look at the financial barriers, noting that professional digital marketing services in the Philippines typically demand monthly retainer fees ranging from ₱25,000 to ₱200,000. For micro-enterprises with limited asset sizes, these costs are entirely unsustainable. Research from MDPI (2024) emphasizes that young entrepreneurs specifically struggle with market conditions where high-frequency content scheduling is required to maintain visibility in saturated digital marketplaces. This leads to "time poverty," where business owners who assume multifarious roles (as product developers and customer service agents) lack the cognitive bandwidth for strategic content management. TR Digital Services (2025) argues that inconsistent posting schedules are a primary cause of low engagement, yet MSMEs lack the tools to automate this process affordably.



Sociolinguistics: The Taglish Identity in Digital Branding

A critical challenge in Philippine digital marketing is the linguistic nuance of the audience. Globalization Partners (2024) describes "Taglish" as a linguistic blend that defines the modern Filipino identity, fusing Tagalog and English into a seamless code-switching phenomenon. Authentic digital branding in the Philippines requires mastery of this nuance to foster trust.

However, generating such content through AI presents technical hurdles. Absar (2025) notes that fine-tuning Large Language Models (LLMs) for Part-of-Speech (POS) tagging in code-switched contexts is significantly more complex than in monolingual data. Traditional AI models often suffer from "Linguistic Negative Transfer," a state identified by Novalins (2025) and POEditor (2024) where models trained primarily on high-resource languages (English) produce "robotic" or culturally sterile outputs when attempting low-resource languages or dialects.

Research by Reyes and Fajardo (2021) on building Filipino word corpora suggests that automatic labeling and segmentation are essential for cultural accuracy. Without this accuracy, the Journals Index Copernicus (2023) warns that businesses face "serious challenges" in resonance, as awkward phrasing can alienate local consumers. Xu (2022) quantifies this risk, suggesting that 24% of consumers would stop their interest in a purchase due to cultural inaccuracies or references that feel "foreign".

Theoretical Foundation: Perceived Brand Localness (PBL)

The effectiveness of Kawayan AI is anchored in the marketing theory of Perceived Brand Localness (PBL). PBL is defined as the extent to which a brand is recognized as an authentic icon or cultural symbol of the local community, embodying its



values and aspirations. According to ResearchGate (2023), high PBL is a key driver of brand trust and purchase intention in developing countries across Asia.

From a Social Identity Theory perspective, PBL creates a sense of "in-group membership". When a consumer perceives a brand as "local" (even if the underlying technology is global), they anchor themselves within that brand's identity, leading to stronger consumer-brand identification. Livermore and Adams (2025) argue that the path to achieving this localness in the AI era is through Culturally Intelligent Prompting, where AI is guided to think beyond raw data and focus on contextual "vibe" and cultural markers (e.g., references to "Pandesal" or "Ube"). This theoretical alignment proves that for Philippine MSMEs, "Taglish" automation is not just a convenience, but a strategic necessity for building legitimacy.

Technical Enablers: Frontier AI and Automated Infrastructure

The realization of a localized content platform depends on the integration of frontier-class technological artifacts. Jasper (2026) has demonstrated the general feasibility of AI marketing platforms, but the specific requirements of the Philippine MSME sector necessitate more localized tools.

The current study leverages the Google Gemini API (specifically versions 3.1 and Gemma 4 released in April 2026) for its advanced reasoning and agentic workflows. Google (2026) notes that Gemma 4 models are "byte for byte" the most capable open models, allowing for complex logic and multi-step planning, which is essential for executing Batch Ideation Generation—creating a month's worth of content in a single transaction.

For operational logistics, the system integrates two critical APIs:



1. **Xendit SDK:** This serves as the financial bridge. As noted in Xendit (2026) documentation, it is the first gateway to offer real recurring subscription payments via local e-wallets like GCash and Maya. This is vital because many Filipino MSME owners do not possess international credit cards.
2. **bundle.social API:** This unified infrastructure streamlines distribution to 14+ platforms, handling "OAuth nightmares" and platform-specific quirks (e.g., TikTok image auto-scaling). This allows MSMEs to "write once, post everywhere," directly addressing the time constraints identified in earlier sections.

Synthesis of the Reviewed Literature

The reviewed literature confirms a critical "Operational Void" in the Philippine MSME ecosystem. While the digital economy is booming , 75% of businesses are excluded due to the high cost of traditional agencies and the linguistic sterility of general AI tools.

The synthesis of these studies establishes that:

- **Macro-Context:** MSMEs are the pillar of the economy but lack digital tools.
- **Technical Problem:** "Linguistic Negative Transfer" makes standard AI robotic for "Taglish" markets.
- **Theoretical Solution:** Increasing Perceived Brand Localness (PBL) through culturally intelligent prompting can restore consumer trust and drive a 24% increase in purchase intent.



- **Artifact Feasibility:** The combination of Gemini's reasoning, Xendit's localized billing, and bundle.social's distribution makes a low-cost, high-impact solution technologically possible.

By bridging these gaps, Kawayan AI serves as an innovative technological artifact that fulfills the mandates of SDG 8 and 9 and the Philippine Development Plan 2023-2028, ensuring that the digital transformation of the country is inclusive and culturally authentic.



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CHAPTER 3

METHODOLOGY

This section outlines the research methods and systematic procedures utilized in the development, implementation, and evaluation of the Kawayan AI platform. The methodology is structured to align with the Design and Development Research (DDR) paradigm, integrating both technical system construction and empirical validation. The chapter details the research design, development framework, modeling artifacts, evaluation instruments, and statistical treatments employed to ensure that the system is both functionally effective and empirically validated within the context of Philippine MSMEs.

The Design and Development Research (DDR) Framework

The Design and Development Research (DDR) paradigm serves as the foundational methodological framework guiding the creation and validation of the Kawayan AI platform. This paradigm is selected due to the nature of the research problem, which extends beyond descriptive analysis and requires the systematic development of a technological artifact capable of addressing real-world operational inefficiencies in MSME digital marketing.

Unlike purely observational methodologies, DDR mandates a dual-phase process consisting of:

1. **Development Phase** – focused on the structured creation of the system artifact
2. **Evaluation Phase** – focused on empirical validation of system effectiveness



The Kawayan AI platform directly addresses a critical operational gap among MSMEs: the inability to consistently produce culturally relevant, high-quality marketing content due to time constraints, financial limitations, and lack of technical expertise. As such, DDR is particularly appropriate because it enables the study to move beyond theoretical analysis and produce a validated, deployable solution.

A key requirement of the DDR framework is **multi-dimensional validation**, which ensures that the developed system is not only functional but also meaningful in practical application. For Kawayan AI, validation is achieved across three domains:

- **Technical Validation** - ensuring system functionality and stability
- **User Acceptance Validation** - measured through TAM constructs (PU & PEOU)
- **Operational Validation** - demonstrated through efficiency gains (e.g., reduced content creation time)

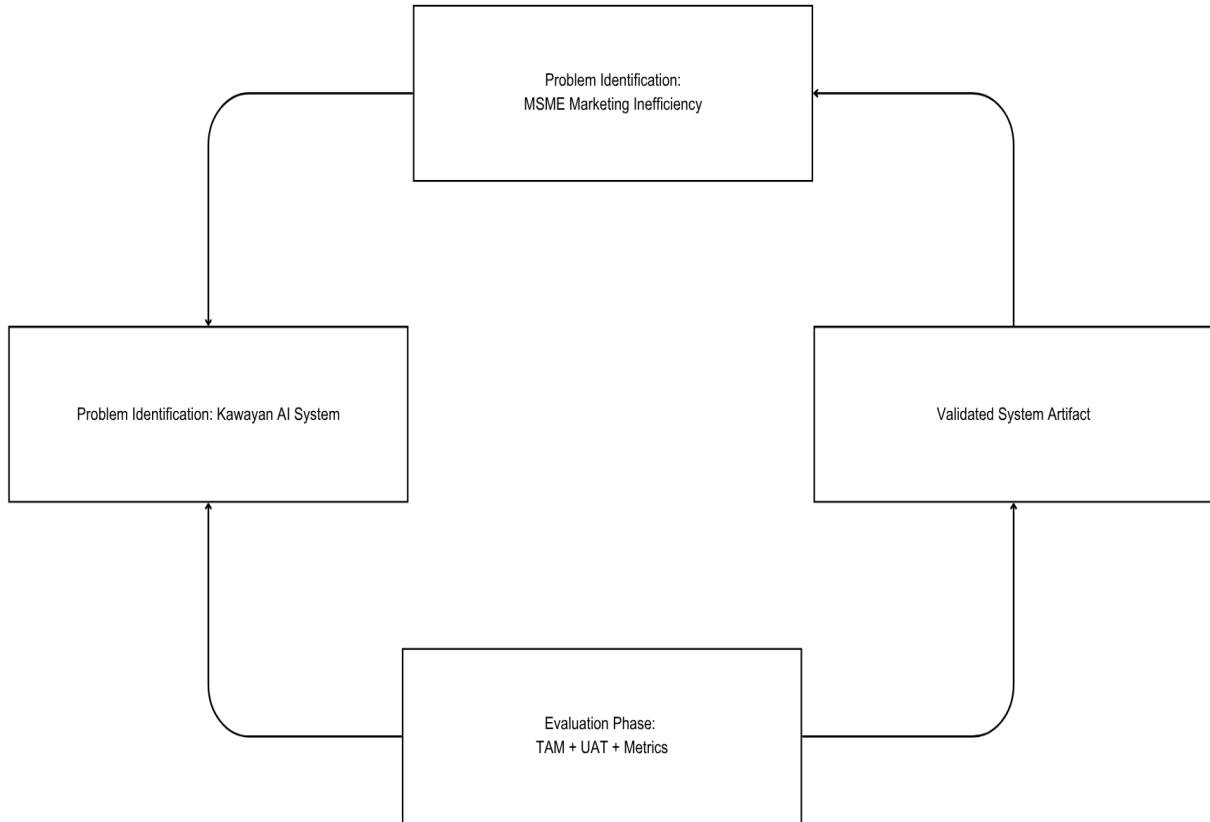
Figure 4. DDR Framework Cycle

Figure 4 illustrates that the DDR cycle begins with the identification of inefficiencies in traditional MSME content creation workflows, particularly the reliance on manual ideation and inconsistent branding. This leads to the structured development of Kawayan AI, which integrates AI-driven content generation with culturally intelligent prompting. The evaluation phase then validates whether the system successfully improves efficiency, usability, and perceived value.

The iterative nature of DDR ensures that insights gained during evaluation are reintegrated into subsequent development cycles, allowing continuous refinement of the system.



Development Methodology: Iterative Prototyping Model (IPM)

To operationalize the Development Phase of the DDR framework, the study adopts the **Iterative Prototyping Model (IPM)** as its software development lifecycle. This model is particularly suitable for systems requiring continuous user feedback and progressive refinement, especially in environments characterized by diverse user capabilities such as MSMEs.

The IPM is structured into four cyclic phases:

1. Planning and Requirements Analysis
2. System Design and Prototyping
3. Development and Implementation
4. Testing and Evaluation

Figure 5. IPM Cycle

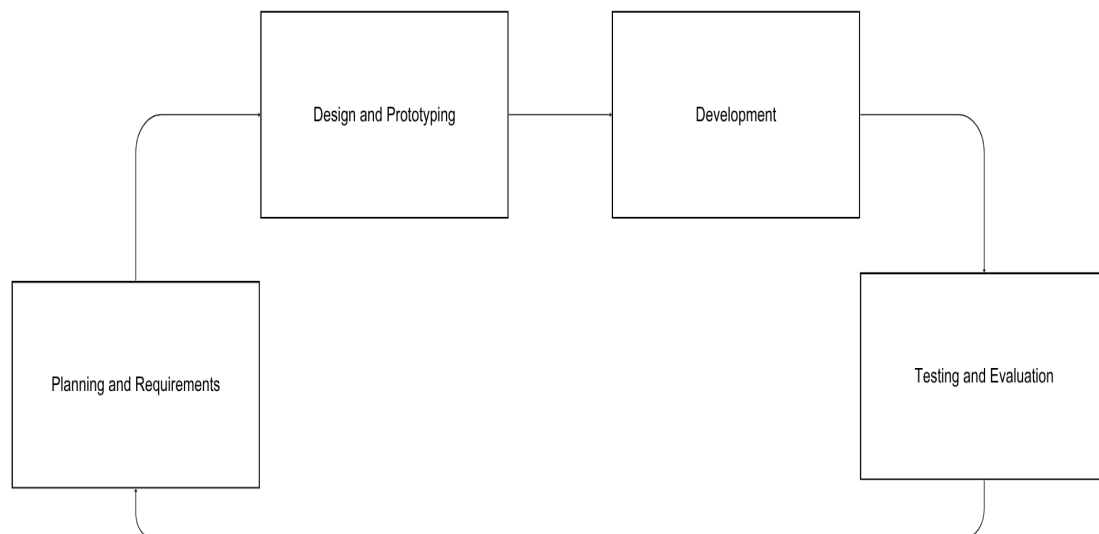




Figure 5 illustrates the Iterative Prototyping Model (IPM) used as the system development methodology. It highlights the continuous cycle of planning, design, development, and testing, ensuring that the Kawayan AI platform evolves through repeated refinements based on user feedback.

- **Phase 1: Planning and Requirements Analysis**

- This phase involves identifying the key operational constraints faced by MSMEs, including:
 - Limited financial resources for marketing services
 - Inconsistent content creation processes
 - Lack of cultural alignment in generic AI outputs
- Functional requirements such as Business Registration and Verification, Document Upload, Verification Status Display, Batch Ideation Generation, Content Scheduling, and Growth Analytics Dashboard are defined alongside non-functional requirements such as low-bandwidth compatibility, system scalability, data privacy, and secure document handling.

- **Phase 2: System Design and Prototyping**

- The system architecture is conceptualized during this phase, including:
 - Frontend user interface for MSMEs
 - Business Verification Module for registration, document upload, and verification status tracking
 - Automation core responsible for AI processing



- Admin Verification Queue for reviewing submitted business documents
- Integration with external APIs (e.g. content generation, scheduling, payment systems)
- **Phase 3: Development and Implementation**
 - This phase involves the actual coding and integration of system components, including:
 - User registration and verification status handling
 - AI Prompt Engineering Pipeline
 - Content generation modules
 - Scheduling automation
 - Dashboard analytics
- **Phase 4: Testing and Evaluation**
 - The system undergoes iterative testing to ensure:
 - Functional correctness
 - Usability and accessibility
 - Performance under varying network conditions
 - Correct account validation status display and access control behavior
 - Feedback gathered during this phase is used to refine the system in subsequent iterations.

Research Approach: Mixed-Methods Strategy

The study adopts a **mixed-methods research approach**, integrating both **quantitative** and **qualitative** techniques to achieve a comprehensive evaluation of the



Kawayan AI platform. This approach is particularly appropriate for Design and Development Research (DDR), as it allows simultaneous assessment of both system performance and user perception.

The **quantitative component** focuses on measuring user acceptance through the Technology Acceptance Model (TAM), specifically evaluating Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) using structured survey instruments. These metrics provide measurable indicators of system effectiveness and adoption potential.

The **qualitative component**, on the other hand, is derived from user feedback during User Acceptance Testing (UAT), where participants provide insights into their experience, challenges encountered, and perceived improvements. This allows the study to capture contextual nuances that cannot be fully quantified through numerical data alone.

The integration of both approaches ensures that the evaluation of Kawayan AI is both statistically grounded and contextually rich, strengthening the validity of the study's findings.

System Modeling and Design Artifacts

To ensure a comprehensive representation of the Kawayan AI system, the study utilizes multiple system modeling techniques. These artifacts provide both conceptual and technical perspectives of the system, enabling clearer visualization of its structure, behavior, and data flow.

The modeling techniques include:

- Use Case Diagram (Functional perspective)
- Class Diagram (Structural perspective)
- Sequence Diagram (Behavioral perspective)



- Data Flow Diagram (Process perspective)

These diagrams also include the account validation process as a supporting system function, ensuring that MSME users can be reviewed before accessing the platform's major content generation and scheduling features.

System Architecture

The Kawayan AI platform follows a modular and scalable architecture composed of interconnected components that facilitate seamless interaction between users, AI services, and system databases.

The architectures consists of:

- **Client Interface** - Web-based interface for MSME users, including account registration, business validation status display, content generation, scheduling, and analytics access
- **Account Validation Module** - Handles submitted business information, uploaded registration documents, and verification status updates
- **Automation Core** - Handles AI processing and business logic
- **AI Prompt Engine** - Manages structured prompt generation
- **External APIs** - Includes Google Gemini API, scheduling API, and payment gateway
- **Database Layer** - SQLite database for multi-tenant data storage, including user records, verification status, generated content, schedules, and analytics data
- **Admin Dashboard** - Provides monitoring, analytics, subscription oversight, and basic account review controls

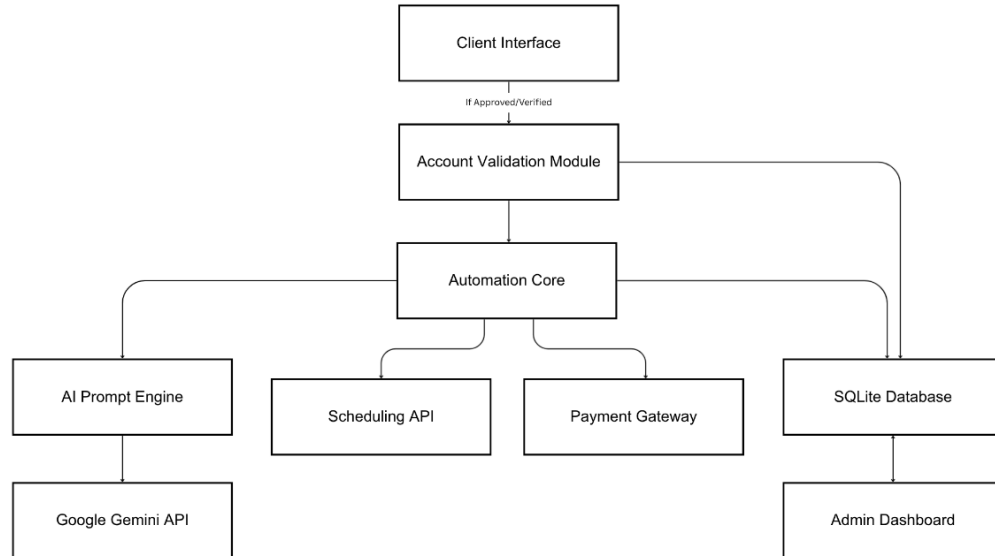
Figure 6. System Architecture

Figure 6 illustrates the system architecture of the Kawayan AI platform. It presents the interaction between the client interface, account validation module, automation core, AI prompt engine, external APIs, database, and admin dashboard. The diagram demonstrates how user inputs are processed through AI services and stored for content generation, scheduling, analytics, and account status management.

Functional Scope

The functional scope of the system is represented through a Use Case model, which identifies the interactions between the user, administrator, and the system. In addition to the core AI content generation and scheduling features, the use case model includes account registration and business validation as supporting functions before full platform access.

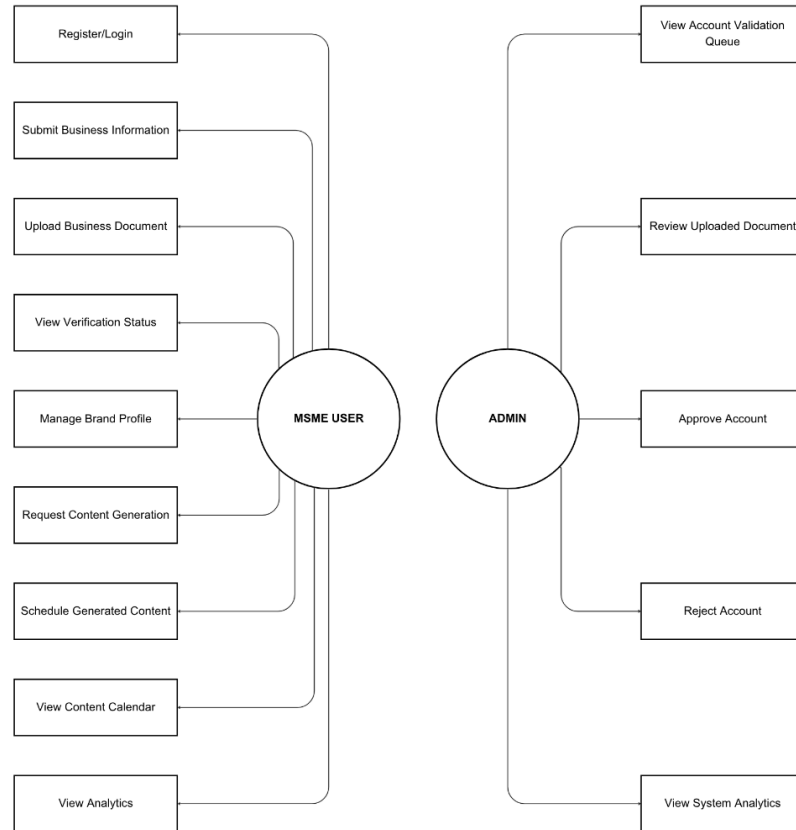
Figure 7. Use Case Diagram

Figure 7 illustrates the use case diagram of the Kawayan AI system. It presents the interactions between the MSME user, the admin, and the system. The MSME user can register or log in, submit business information, upload a business document, view verification status, manage the brand profile, request content generation, schedule generated content, view the content calendar, and access analytics. These functions represent the main user-side features of the platform, from account validation to AI-assisted content management.

The admin actor is responsible for supporting platform oversight and account validation. Through the system, the admin can view the account validation queue, review uploaded business documents, approve or reject account verification requests, and view



system analytics. This ensures that business verification is included as a supporting function while maintaining the platform's primary focus on content generation, scheduling, and analytics for MSME digital visibility.

Structural Model

The structural components of the system are represented through a class diagram, illustrating the relationships between key entities.

Figure 8. Class Diagram

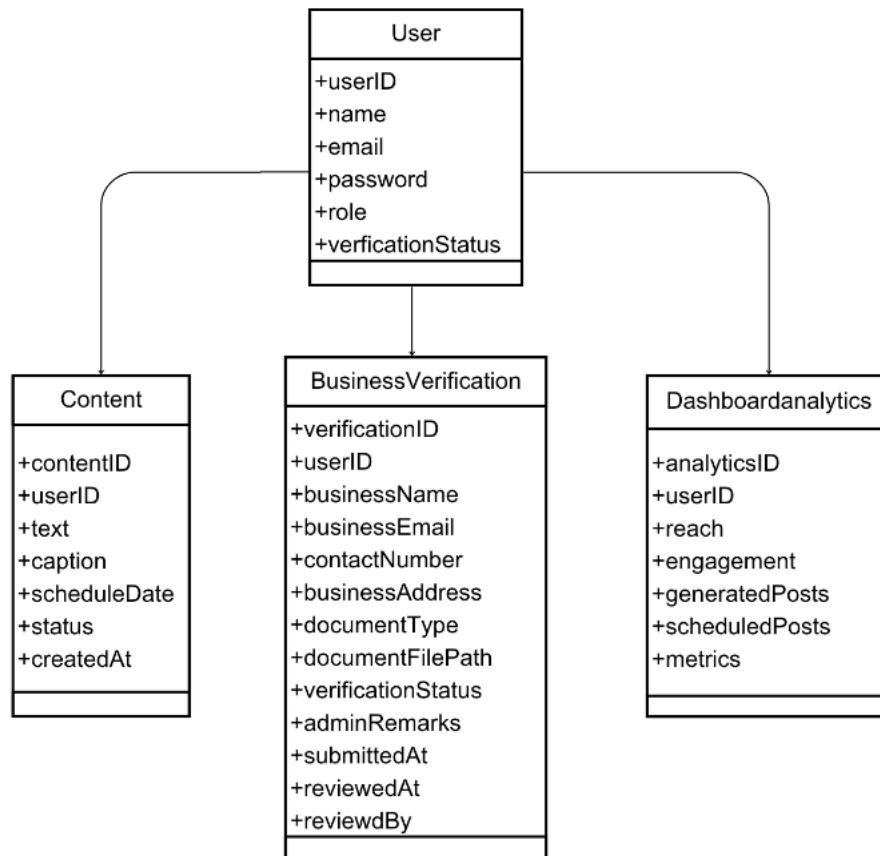


Figure 8 illustrates the class diagram representing the structural design of the system. It shows the relationships between key entities such as the User, Content, Dashboard, and BusinessVerification, including their attributes and associations within



the Kawayan AI platform. The BusinessVerification class supports account validation by storing submitted business information, uploaded document details, and verification status.

Behavioral Model

The sequence diagram illustrates the interaction flow during content generation.

Figure 9. Sequence Diagram

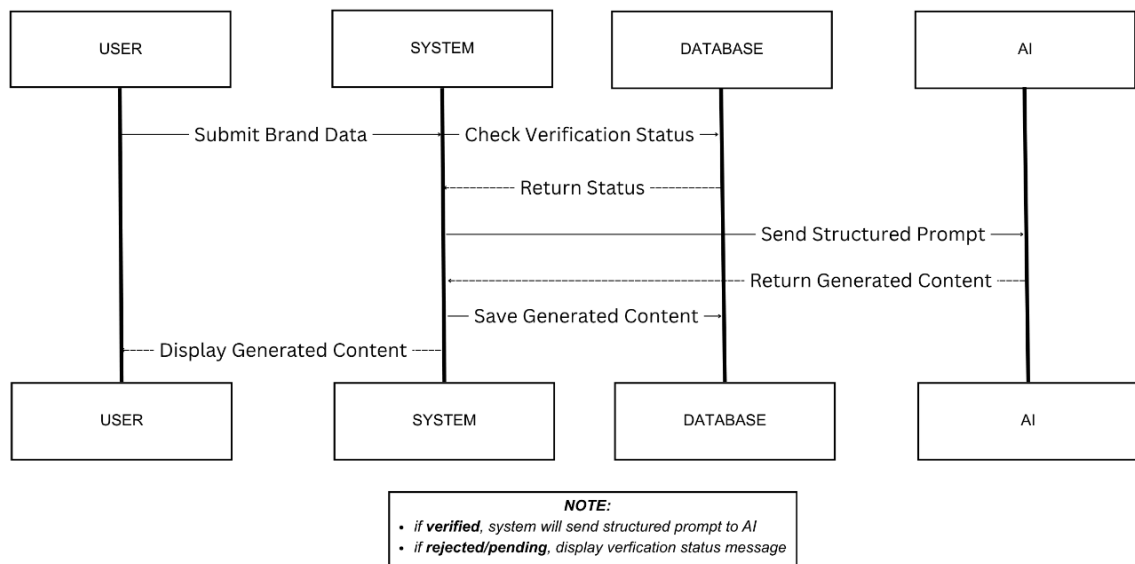


Figure 9 illustrates the sequence diagram of the Kawayan AI content generation process. The user submits brand data to the system, and the system checks the user's verification status from the database. If the user is verified, the system sends a structured prompt to the AI service, receives the generated content, saves the output in the database, and displays the generated content to the user. If the account is still pending or rejected, the system displays the appropriate verification status message instead of proceeding with content generation.

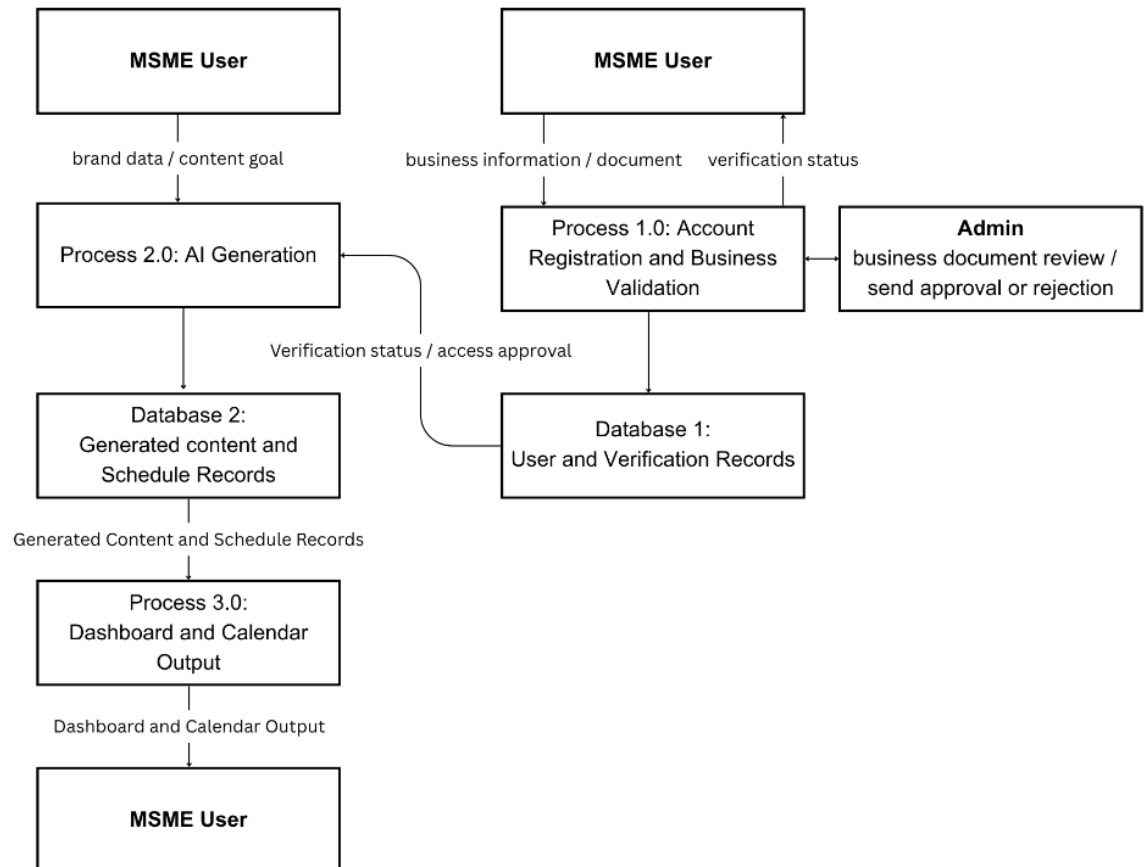
Data Flow Diagram**Figure 10. DFD Level 1**

Figure 10 presents the Level 1 Data Flow Diagram of the Kawayan AI platform. It shows how MSME users submit business information and uploaded documents for account registration and business validation, while the admin reviews the submitted document and sends an approval or rejection decision. The user and verification records are stored in the database and used to determine whether the user may proceed to AI generation. Once approved, the MSME user submits brand data and content goals, which are processed by the AI generation module. The generated content and schedule



records are stored and later displayed to the user through the dashboard and content calendar output.

Locale and Scope of the study

The study is conducted in Parañaque City, Philippines, a highly urbanized area characterized by a diverse and active population of Micro, Small, and Medium Enterprises (MSMEs). The selection of this locale is strategic due to its strong adoption of digital platforms for business operations, particularly social media-based marketing, making it an appropriate environment for evaluating the Kawayan AI system.

Parañaque City presents a representative sample of urban MSMEs in the Philippines, where business owners often rely on digital tools to maintain competitiveness in a rapidly evolving market. However, despite high digital engagement, many MSMEs continue to face challenges in producing consistent, culturally relevant, and high-quality content due to limitations in time, budget, and technical expertise.

The scope of the study is limited to the design, development, and evaluation of the Kawayan AI platform, specifically focusing on its ability to:

- Generate culturally intelligent and localized marketing content
- Facilitate automated content scheduling
- Provide basic analytics through a dashboard interface

The study does not cover financial performance metrics, long-term business growth outcomes, or comparative analysis with other AI platforms. Instead, it focuses on evaluating system usability, efficiency, and user acceptance within the defined locale.

Study Population and Target Users

The study population consists of Micro, Small, and Medium Enterprise (MSME) owners and operators who actively utilize social media platforms as part of their



marketing strategies. These individuals represent the primary target users of the Kawayan AI system, as the platform is specifically designed to address their operational challenges in digital content creation.

The target users are characterized by:

- Direct involvement in managing business social media accounts
- Basic familiarity with digital tools and platforms
- Limited access to professional marketing services

These users often perform multiple roles within their businesses, including marketing, operations, and customer engagement, which increases the need for automated tools that can streamline repetitive tasks such as content creation.

By focusing on MSME owners and operators, the study ensures that the evaluation of the system is grounded in real-world use cases, thereby increasing the practical relevance and applicability of the research findings.

Sampling Technique and Sample Size

The study employs a purposive sampling technique, a non-probability sampling method that selects participants based on specific criteria relevant to the research objectives. This approach is appropriate for the study as it ensures that only individuals with direct experience in managing digital marketing activities are included.

The selection criteria for respondents include:

- Ownership or active management of an MSME
- Use of social media platforms for business promotion
- Willingness to participate in system testing and evaluation

A total of 40 respondents will be selected for participation in the study. This sample size is justified based on commonly accepted standards in undergraduate IT



research, where a range of 30 to 50 respondents is considered sufficient for descriptive statistical analysis and system evaluation.

The chosen sample size balances data reliability and practical feasibility, allowing for meaningful analysis of user perceptions while remaining manageable within the study's time and resource constraints.

Research Instruments and Evaluation Metrics

The study utilizes a TAM-based survey questionnaire as the primary instrument for evaluating user acceptance of the Kawayan AI platform. The questionnaire is designed using a 5-point Likert scale to capture the perceptions of respondents regarding the system's usefulness and ease of use.

The evaluation metrics are based on the core constructs of the Technology Acceptance Model (TAM), including:

- **Perceived Usefulness (PU):** Measures the extent to which users believe that the system enhances their productivity and effectiveness in content creation and marketing.
- **Perceived Ease of Use (PEOU):** Measures the extent to which users find the system easy to understand, navigate, and operate.

In addition to TAM constructs, the study incorporates system performance metrics, including:

- Content generation speed
- System responsiveness
- Stability during use



These metrics provide a more comprehensive evaluation by combining subjective user perceptions with objective system performance indicators, ensuring a well-rounded assessment of the platform.

Validity and Reliability

To ensure the credibility and accuracy of the research findings, the study implements both validity and reliability measures for the survey instrument.

Validity

The questionnaire will undergo expert validation, where IT professionals and academic faculty members review the instrument to ensure that:

- The questions are aligned with the TAM constructs
- The items are clear, relevant, and appropriate for the target respondents
- The instrument accurately measures the intended variables

This process ensures content validity, confirming that the instrument effectively represents the concepts being studied.

Reliability

To assess the internal consistency of the questionnaire, the study will use Cronbach's Alpha, a widely accepted statistical measure for reliability testing.

A Cronbach's Alpha value of:

- 0.70 or higher will be considered acceptable
- 0.80 or higher will be considered good
- 0.90 or higher will indicate excellent reliability

This ensures that the responses collected from participants are consistent and dependable, strengthening the overall validity of the study's conclusions.



Data Gathering Procedure

Figure 11. Data Gathering Process

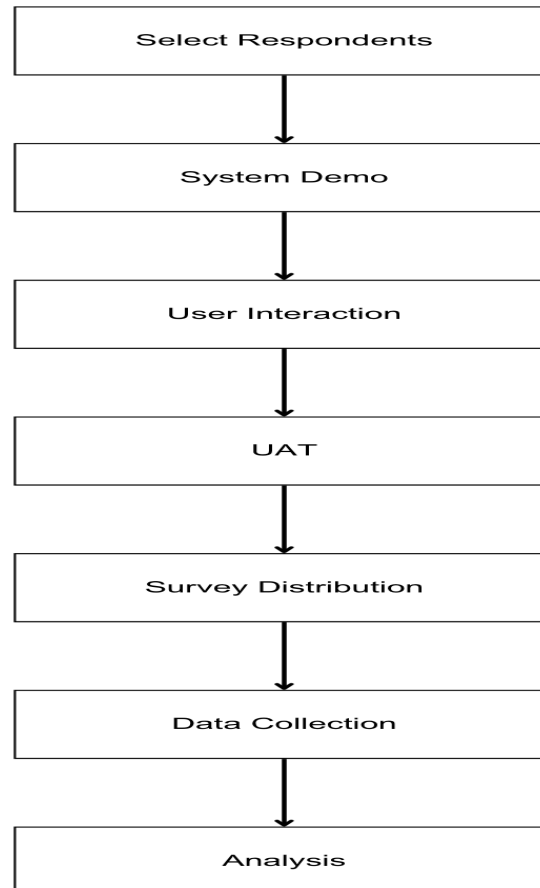


Figure 11 illustrates the data gathering procedure followed in the study. It outlines the sequential process from respondent selection to system demonstration, user interaction, survey distribution, and final data analysis.



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